Tournament anti-Sidorenko orientations of trees with at most five branch vertices

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Abstract

A finite tree with at most five vertices of degree at least three admits an orientation whose homomorphism density into any tournament is at most the random-tournament benchmark. Degrees, arm multiplicities and positive subdivision lengths are unrestricted, and a single orientation, chosen from the tree alone, satisfies the stronger kernel inequality for every finite probability space and every real skew kernel of absolute value at most one. For four branch vertices we convert terminal fourth moments into weighted second moments across a connector; for five we allocate reference arms in the star and fork skeletons and join the two halves of the path skeleton through compatible fractional moments. Separate arguments cover a central singleton arm and a unit outer connector. The kernel inequality also yields an entropy bound that holds for arbitrarily correlated vertex labels.

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1 Introduction

For an oriented graph D and a finite tournament $\mathcal{T}$, let $\hom(D,\mathcal{T})$ be the number of arc-preserving vertex maps from D to $\mathcal{T}$; the maps need not be injective. An orientation D of a tree U is *tournament anti-Sidorenko* if

$$\hom(D,\mathcal{T}) \le 2^{-e(U)}|V(\mathcal{T})|^{|V(U)|} \tag{1.1}$$

for every nonempty finite loopless tournament. The factor $2^{-e(U)}$ is the limiting density of D in a uniformly random tournament. Thus (1.1) bounds the number of maps from D into any tournament by this random benchmark. Fox, Himwich, Mani and Zhou conjectured that every tree admits such an orientation [3, Conjecture 1.12]. We prove their conjecture for trees with at most five branch vertices.

Here a *branch vertex* of U is a vertex of degree at least three, and

$$b(U) = |\{v \in V(U) : \deg_U(v) \ge 3\}|, \qquad e(U) = |E(U)|.$$

All degrees refer to U.

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Theorem 1.1. *Every nonempty finite tree U with $b(U) \le 5$ has an orientation $D = D(U)$ satisfying (1.1) for every nonempty finite loopless tournament $\mathcal{T}$.*

We prove a kernel form of the theorem, in which one orientation serves every host. For a nonempty finite probability space (X, μ) and a real skew kernel $K : X^2 \to [-1, 1]$, put

$$Z_D(K, \mu) = \mathbb{E}_{\mu^{V(U)}} \prod_{u \to v \in E(D)} (1 + K(x_u, x_v)). \tag{1.2}$$

All labels are independently sampled from the same measure; coincident labels are included.

Theorem 1.2. *For every nonempty finite tree U with $b(U) \le 5$, there is an orientation $D = D(U)$ of all its edges such that*

$$Z_D(K, \mu) \le 1 \tag{1.3}$$

for every nonempty finite probability space and every real skew kernel bounded in absolute value by one.

Related results.

Sidorenko's conjecture bounds the homomorphism density of a bipartite graph from below in terms of its edge density [9]; the tournament problem seeks an orientation with a universal upper bound at the random benchmark. Table 1 lists the tree classes covered by the relevant results. These classes overlap only partly: arbitrary subdivisions produce many vertices of degree two, whereas a bound on $b(U)$ restricts neither the degrees nor the multiplicities of equal branches.

Table 1: Tree classes covered by the closest prior results and by this paper.

Work	Scope
Fox–Himwich–Mani–Zhou [3, Proposition 1.13]	Trees with at most one vertex of even degree.
Chen–Clemen–Noel [2]	Every three-legged spider, with arbitrary leg lengths.
He–Mani–Nie–Tung–Wei [6, Theorems 1.3 and 1.4]	Isomorphic-branch attachment and every caterpillar.
Theorem 1.1	Every tree with at most five original vertices of degree at least three, with unrestricted degrees, multiplicities and positive subdivision lengths.

Chen and Lin prove path anti-Sidorenko inequalities when the length is large relative to the number of direction flips [1]. Paths enter our argument in a different role, as connectors that carry rooted messages between branch vertices; their lengths are arbitrary, including one.

Proof strategy.

The proof separates the assembly of a tree from the local moment estimates. Table 2 gives the route through the assembly; the technical appendix proves the inputs listed there.

For four branch vertices the terminal fourth moments close the path P_4 and the star $K_{1,3}$. For five, the star and fork need one existing central arm as a reference, while the path uses $I_q(V) = \mathbb{E}[q^{-1/3}|V|^{8/3}]$, with $q = 1 - K\mathbf{1}$. After the two structural path alternatives, both side messages satisfy this bound with the same q; the three weight powers cancel in the central Hölder inequality. The arm allocation, rather than a restriction on the connector lengths, distinguishes these assemblies.

Appendix A gives the shared scalar calculus, and Appendix B proves one terminal estimate in full. Appendix L is the entry point to the remaining direction tables: it separates reference, ordinary and outlet inputs, whose directions and weights differ. The local proofs retain the short-length exceptions before treating the finite tails.

Table 2: The proof of Theorems 1.1 and 1.2. Each row uses the preceding structural reductions and the indicated local estimates.

Stage	Argument or required input	Location
Path interface	Weighted second moments at two roots control every positive connector length.	Lemma 2.1
Four branches	Pair equal arms; convert terminal fourth moments to the second moments needed by the interface.	Sections 4–7
Five-branch reduction	Reduce to four branches or to P_5, $K_{1,4}$ and the fork T_5.	Lemma 8.1
Star and fork	Allocate disjoint arms to simple-reference fourth, ordinary eighth and compound-reference second moments.	Section 10; Propositions 9.1–9.3
Path alternatives	First treat a central singleton and a unit off-center arm whose receiving connector has length one.	Section 11
Remaining path	Give both sides the same inverse-weighted boundary estimate; close by Hölder at the center.	Proposition 9.4; (11.1)
Conclusion	Fix all choices from the tree, restore equal pairs, and pass from kernels to tournaments.	Section 12

2 Kernel preliminaries and rooted messages

Delete zero-mass atoms and use the real inner product $\langle u, v\rangle = \mathbb{E}(uv)$ on $L^2(\mu)$. Define

$$Ju = (\mathbb{E}u)\mathbf{1}, \quad A = J+K, \quad A^* = J-K, \quad g = K\mathbf{1}, \quad d = \mathbb{E}g^2, \quad p = 1+g, \quad q = 1-g. \tag{2.1}$$

All operators and adjoints are taken with respect to this measure. We also write $r = Kg$. Write $T = A^*$. Both A and T have nonnegative kernels. Since $K(x,x) = 0$ and $\mu(x) > 0$, we have $p(x), q(x) \geq \mu(x) > 0$.

The row constraint is

$$|r(x)| \leq [1 - g(x)^2]/2 = p(x)q(x)/2. \tag{2.2}$$

For $a_y = K(x,y)$, antisymmetrization gives $r(x) = \frac{1}{2}\mathbb{E}_{y,z}[(a_y - a_z)K(y,z)]$. The scalar inequality $|u-v| \leq 1-uv$ on $[-1,1]^2$ follows by squaring, since the difference is $(1-u^2)(1-v^2) \geq 0$. Taking expectations proves (2.2).

For an orientation D of a rooted tree, its message is the conditional integral of the product in (1.2), with the root label fixed and all other labels integrated independently.

For $n \geq 0$ put $f_n = A^n\mathbf{1}$, $\bar{f}_n = (A^*)^n\mathbf{1}$, $\mu_n = \mathbb{E}f_n$ and, for $n \geq 1$, $h_n = A^*f_{n-1}$. At a rooted arm of length n, these are the actual messages of, respectively, all edges pointing away from the root, all edges pointing toward the root, and the first edge pointing toward the root with every remaining edge pointing away along the arm. In particular, $f_1 = p$ and $h_1 = q$.

We use the constants and moments

$$\theta = \frac{13}{32}, \qquad C_* = \frac{101}{50}, \qquad \delta_j = \left\|K^j\mathbf{1}\right\|_2^2, \qquad S_n = \sum_{j=0}^{n} K^j\mathbf{1}. \tag{2.3}$$

Appendix D proves $\|K\|_{2\to 2} \leq 2/\pi$, $d \leq 1/3$, $\delta_{j+1} \leq \theta\delta_j$, $\mu_n \leq 1$, and the positive convolution identity

$$S_n = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad d - \delta_2 \leq \beta_j \leq d \quad (j \geq 3). \tag{2.4}$$

The elementary identity $A + A^* = 2J$ gives

$$f_n + h_n = 2\mu_{n-1}, \qquad 0 < f_n h_n \leq \mu_{n-1}^2 \leq 1. \tag{2.5}$$

At a root, pair equal arm lengths maximally and use f_n, h_n on each pair; let E_0 be the actual product of the pair messages, so $0 < E_0 \le 1$. The remaining lengths are distinct and are exactly those of odd multiplicity. Since $E_0 \le 1$, an upper bound may drop E_0.

For an even arm list, its *positive canonical orientation* pairs the remaining distinct lengths adjacently, assigning $h_a f_b$ to a pair $a < b$. For an odd list, retain the shortest residual arm as an outward anchor f_a and pair the rest in this way. Write the odd message as $f_a R$, where R includes all equal and distinct pairs. The *negative canonical orientation* reverses all the corresponding edges. A universal assertion at $-K$ proves the reversed one.

For later use set

$$N_+(H) = \mathbb{E}[pH^2], \qquad N_-(H) = \mathbb{E}[qH^2].$$

Lemma 2.1 (Weighted path interface)**.** *For every integer $L \ge 1$ and all real H, G,*

$$|\langle H, A^L G\rangle| \le \sqrt{N_+(H) N_-(G)}, \qquad N_-(AG) \le N_-(G).$$

The reversed statements hold after interchanging A, T and p, q.

Proof. The nonnegative edge measure $(1 + K(x,y))\, d\mu(x) d\mu(y)$ has marginals $p\mu, q\mu$. Cauchy–Schwarz on this measure proves the first inequality for $L = 1$. Row Cauchy–Schwarz gives $(AG)^2 \le pA(G^2)$, and hence

$$N_-(AG) \le \mathbb{E}[pqA(G^2)] = \mathbb{E}[T(pq)G^2] \le \mathbb{E}[qG^2],$$

because $pq = 1 - g^2 \le 1$ and T preserves order. Iterate $L-1$ times and apply the one-edge inequality. □

Lemma 2.2 (Rooted spider boundary)**.** *Every finite arm multiset admits an orientation, fixed independently of the kernel and measure, with actual message F satisfying $N_-(F) \le 1$.*

The direction for each residual arm list, and the proof, are given in Appendix D.5. Reversal gives the corresponding N_+ statement.

Table 3: Notation used in the moment calculations.

Symbol	Meaning
g, r	$K\mathbf{1}$ and Kg, both with the original measure.
r_0	The centered function $QKg = Kg + d\mathbf{1}$, used only in Appendix E.
d, δ_j	$d = \mathbb{E}g^2 = \delta_1$ and $\delta_j = \|K^j\mathbf{1}\|_2^2$. Ordinary powers such as d^2 remain powers of d.
$f_n, \bar{f}_n, h_n$	Outward, inward, and first-edge-inward arm messages, respectively.
β_j, ω_j	Positive convolution coefficients, and local envelope or propagation losses; neither is a path function.
ξ	A scalar score coordinate in a polynomial comparison: $p(x) = \xi$, $q(x) = 2 - \xi$.
n_T	Number of vertices of the host tournament.

3 The four-branch foundation

The five-branch argument uses the following four-branch result whenever a reduction removes one branch vertex.

Theorem 3.1. *For every nonempty finite tree U with $b(U) \le 4$, there is an orientation $D = D(U)$ of all its edges such that (1.1) holds for every nonempty finite loopless tournament.*

Theorem 3.2. *The orientation in Theorem 3.1 can be chosen so that $Z_D(K,\mu) \le 1$ for every finite probability space and every real skew kernel bounded in absolute value by one.*

Appendix C proves the assertion for at most three branch vertices. Section 4 reduces to the two skeletons, Section 5 supplies the terminal estimates and the internal side tables; Sections 6 and 7 then close the chain and the star. These inputs also serve the five-branch case.

4 The two branching skeletons and a reduction

Theorem C.2 gives every tree with at most three branch vertices one orientation satisfying (1.3) for every finite model, so it remains to treat $b(U) = 4$.

Let B be the four original branch vertices, and let S be the minimal connected subtree of U containing B. Every vertex of degree at least three in S belongs to B, since it also has original degree at least three. Every leaf of S belongs to B, by minimality. Suppressing the degree-two vertices of S outside B leaves a tree on exactly four vertices, which is either a path P_4 or a star $K_{1,3}$ (Figure 1).

Each skeleton edge represents a positive-length path in the original tree, whose interior vertices have degree two and carry no side arms. Every component outside the core is a path attached at a branch vertex, since another branch point there would belong to B, contrary to the choice of S. These components are the *side arms*, and the core paths together with the side arms partition the edges of U.

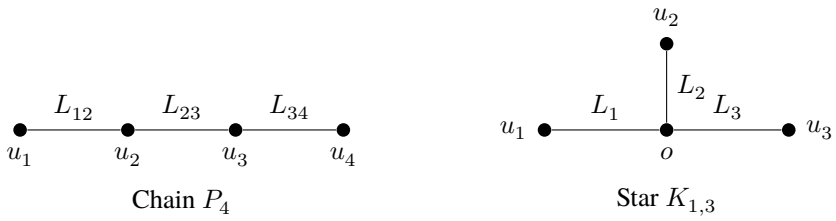


Figure 1: The two skeletons on four branch vertices. Each edge represents its original positive-length connector; side arms are omitted from the drawing.

A leaf of the skeleton is called a *terminal*. It has at least two actual side arms in U. Maximally pair its equal-length arms using (2.5).

Lemma 4.1 (Reducible terminal)**.** *If a terminal has zero or one residual side arm after equal-length pairing, then U has a fixed orientation satisfying* (1.3)*.*

Proof. Orient the equal pairs so that their product is $E_i \le 1$. Removing those paired arms for the comparison leaves the terminal vertex, its remaining arm if any, and all core edges; the terminal then has degree at most two and no other degree has increased. The comparison tree is nonempty, has at most three branch vertices, and carries a fixed orientation from Theorem C.2. Restoring those arms multiplies the nonnegative integrand at the terminal label by $E_i \le 1$, preserving the density bound. □

The reduction applies to both skeletons. We may therefore assume that every terminal has at least two distinct residual lengths. Singleton internal side tables and singleton star-center tables survive the reduction and are estimated in Sections 6 and 7.

5 The complete terminal fourth-moment catalogue

Let Σ be a terminal arm multiset whose distinct residual list has at least two entries, with arbitrary original multiplicities. We use the following local estimate.

Theorem 5.1 (Terminal supply)**.** *For each integer $n \geq 0$ there is an orientation of all the arms, with actual message $F^{+}_{\Sigma,n}$, determined by Σ and n, such that*

$$\mathbb{E}\big[p(T^n F^{+}_{\Sigma,n})^4\big] \leq 1. \tag{5.1}$$

Its complete reversal $F^{-}_{\Sigma,n} = \bar{F}^{+}_{\Sigma,n}$ satisfies

$$\mathbb{E}\big[q(A^n F^{-}_{\Sigma,n})^4\big] \leq 1. \tag{5.2}$$

The parameter n is a prescribed integer determined by the original connector length.

The catalogue below proves the theorem, with analytic estimates in the indicated sections of Appendices H–J.

5.1 Zero-step directions

Every direction in this subsection includes the equal-pair multiplier E_0. For an even residual list, put

$$P = \prod_{i=1}^{t} h_{a_i} f_{b_i}, \qquad a_1 < b_1 < \cdots < a_t < b_t.$$

This direction $F^{+}_{\Sigma,0} = E_0 P$ satisfies $\mathbb{E}[p(F^{+}_{\Sigma,0})^4] \leq 1$. The pair $(1,2)$ is treated directly in Section H.8. Every other even residual list also satisfies a weighted sixth-moment bound, which yields the fourth moment by Hölder against $p\mu$; the complete even catalogue, with its component ranges, is collected at the end of Section H.8, with proofs in Appendix D and Sections H.1–H.7.

For an odd residual list $a < b_1 < c_1 < \cdots < b_t < c_t$, $t \geq 1$, use

$$F^{+}_{\Sigma,0} = \begin{cases} E_0 q f_{b_1} h_{c_1} \displaystyle\prod_{i=2}^{t} h_{b_i} f_{c_i}, & a = 1, \\ E_0 f_a \displaystyle\prod_{i=1}^{t} h_{b_i} f_{c_i}, & a \geq 2. \end{cases} \tag{5.3}$$

The first line is proved in Section H.12, for every number of pairs. The second line is the full reversal of the q-weighted fourth estimates for positive canonical odd messages, with minimum two in Section H.10 and minimum at least three in Section H.11; there the equal-pair arms are reversed together with the residual arms, which is what converts the q weight into the p weight.

5.2 Positive steps: even residual lists

Write the residual list as $a_1 < b_1 < \cdots < a_t < b_t$, $t \geq 1$. The following table gives the residual part of $F^{+}_{\Sigma,n}$ for every $n \geq 1$; multiply each row by its actual E_0.

The row of minimum three uses $f_3 h_{b_1}$, with the first pair reversed.

5.3 Positive steps: odd residual lists

Write

$$a < b_1 < c_1 < b_2 < c_2 < \cdots < b_t < c_t, \qquad t \geq 1.$$

Again the table gives only the residual part and covers all $n \geq 1$. Here $\bar{f}_a = T^a \mathbf{1}$ is a path function, whereas b_i in the ordered list is an arm length.

Table 4: Positive-step terminal directions for even residual lists.

Residual condition	Residual message	Complete proof
$t = 1, (a_1, b_1) = (1, 2)$	qf_2	Section B.1
$t = 1, a_1 = 1, b_1 \geq 3$	qf_{b_1}	Section I.1
$t \geq 2, a_1 = 1$	$\prod_{i=1}^{t} h_{a_i} f_{b_i}$	Section I.6
$t = 1, (a_1, b_1) = (2, 3)$	$h_2 f_3$	Section I.2
$t = 1, a_1 = 2, b_1 \geq 4$	$h_2 f_{b_1}$	Section I.3
$t \geq 2, a_1 = 2$	$\prod_{i=1}^{t} h_{a_i} f_{b_i}$	Section I.8
$a_1 = 3$, any $t \geq 1$	$f_3 h_{b_1} \prod_{i=2}^{t} h_{a_i} f_{b_i}$	Sections I.5, I.7
$a_1 \geq 4$, any $t \geq 1$	$\prod_{i=1}^{t} h_{a_i} f_{b_i}$	Section I.4

Table 5: Positive-step terminal directions for odd residual lists.

Residual condition	Residual message	Complete proof
$a = 1, t = 1, b_1 = 2, c_1 = 3$	$qf_2 h_3$	Section J.1
$a = 1, t = 1, b_1 = 2, c_1 \geq 4$	$qf_2 h_{c_1}$	Section J.2
$a = 1, t = 1, 3 \leq b_1 < c_1$	$qf_{b_1} h_{c_1}$	Section J.3
$a = 1, t \geq 2$	$qf_{b_1} h_{c_1} \prod_{i=2}^{t} h_{b_i} f_{c_i}$	Section J.6
$a = 2, b_1 = 3$, any $t \geq 1$	$\bar{f}_2 f_3 h_{c_1} \prod_{i=2}^{t} h_{b_i} f_{c_i}$	Section J.5
$a \geq 2, b_1 = 4$	$\bar{f}_a f_4 h_{c_1} \prod_{i=2}^{t} h_{b_i} f_{c_i}$	Section J.4
$a \geq 2, b_1 \geq 5$	$\bar{f}_a \prod_{i=1}^{t} h_{b_i} f_{c_i}$	Section J.4

Completeness of the terminal classification.

For an even list the minimum is $1, 2, 3$, or at least four; the first two cases split into one pair and multiple pairs. For an odd list of minimum one, the cases split by the number of pairs and then by the second length. At minimum two the second length is three, four, or at least five; at larger minimum it is at least four. The inward anchor $\bar{f}_a$ is an actual arm with all of its edges directed toward the root.

5.4 Restoration, reversal, and interface length

The proofs in the tables are for the residual message and already allow arbitrary equal-pair factors. Since $0 \leq E_0 \leq 1$ and $P \geq 0$ give $T^n(E_0 P) \leq T^n P$, taking fourth powers and integrating against the nonnegative weight p restores the paired arms without weakening the bound.

Applying (5.1) at $-K$ interchanges T with A, p with q, and the selected arm message with its complete reversal, which proves (5.2).

For a connector of original length L, keeping the edge nearest the internal vertex leaves $n = L - 1$ edges to integrate, so $L = 1$ uses the zero-step catalogue and $L \geq 2$ the positive-step one. In the empty-center star construction of Section 7.4 one terminal is propagated through all L edges.

5.5 The two internal side-table roles

After equal pairing, an internal side table in a chain is of one of two kinds. Call it type R if its residual list is empty or has at least two entries. For an empty list take $V^+ = E_0$, and otherwise use the zero-step catalogue. In either case

$$\mathbb{E}[p(V^+)^4] \leq 1, \qquad V^- = \bar{V}^+, \qquad \mathbb{E}[q(V^-)^4] \leq 1. \tag{5.4}$$

Call the table type S if it has exactly one residual arm, of length $a \geq 1$. Put $C^+ = E_0 f_a$ and $C^- = \bar{C}^+$. The complete singleton estimate is

$$\mathbb{E}[q^5 (C^+)^4] \leq 1, \qquad \mathbb{E}[p^5 (C^-)^4] \leq 1. \tag{5.5}$$

For $a = 1$, $q^5 p^4 = q(pq)^4 \leq q$ proves this directly. The length-two proof is Section G.2, and all lengths $a \geq 3$ are proved in Section G.3.

6 The chain skeleton

Let u_1, u_2, u_3, u_4 be the branch vertices in path order. Write $L_{12}, L_{23}, L_{34} \geq 1$ for their original connector lengths. By Lemma 4.1, the two terminals may be assumed irreducible. Every central-path edge will point from u_2 toward u_3.

6.1 Four half-tree inequalities

The following inequalities account for the weights in the internal catalogue; all factors in each product are evaluated at the same root label.

Lemma 6.1. *For nonnegative functions C, F, G,*

$$N_+(CTF) \leq \sqrt{\mathbb{E}[qC^4]\,\mathbb{E}[pF^4]}, \tag{6.1}$$

$$N_-(CTF) \leq \sqrt{\mathbb{E}[q^5 C^4]\,\mathbb{E}[pF^4]}, \tag{6.2}$$

$$N_-(CAG) \leq \sqrt{\mathbb{E}[pC^4]\,\mathbb{E}[qG^4]}, \tag{6.3}$$

$$N_+(CAG) \leq \sqrt{\mathbb{E}[p^5 C^4]\,\mathbb{E}[qG^4]}. \tag{6.4}$$

Proof. Row Cauchy–Schwarz gives $(TF)^2 \leq qT(F^2)$. Hence

$$N_+(CTF) \leq \mathbb{E}[pqC^2 T(F^2)] \leq \mathbb{E}[C^2 T(F^2)].$$

The edge measure for T has root marginal $q\mu$ and other marginal $p\mu$. Cauchy–Schwarz on that measure proves (6.1). For (6.2), retain the extra factors:

$$N_-(CTF) \leq \mathbb{E}[q^2 C^2 T(F^2)] \leq \sqrt{\mathbb{E}[q(q^2C^2)^2]\,\mathbb{E}[pF^4]}.$$

The first term is exactly $\mathbb{E}[q^5 C^4]$. Complete reversal gives (6.3) and (6.4). □

The merger converts fourth moments into the second moments required by Lemma 2.1.

6.2 Compatible directions for all four role combinations

At the left terminal set $n_1 = L_{12} - 1$ and use Theorem 5.1 to define

$$G_1^+ = T^{n_1} F^+_{1,n_1}, \qquad G_1^- = A^{n_1} F^-_{1,n_1}.$$

Then $\mathbb{E}[p(G_1^+)^4] \leq 1$ and $\mathbb{E}[q(G_1^-)^4] \leq 1$. Use $n_4 = L_{34} - 1$ at the right terminal to obtain $G_4^\pm$ with the same bounds. The superscript at a terminal is chosen according to its adjacent internal role in the following table.

Table 6: Compatible outer directions and half-tree bounds.

Role	Side message	Entire outer path	Half-tree message	Required output
Left R	V_2^-	$u_1 \to u_2$	$H_\ell = V_2^- T G_1^+$	$N_+(H_\ell) \le 1$
Left S	C_2^-	$u_2 \to u_1$	$H_\ell = C_2^- A G_1^-$	$N_+(H_\ell) \le 1$
Right R	V_3^+	$u_3 \to u_4$	$H_r = V_3^+ A G_4^-$	$N_-(H_r) \le 1$
Right S	C_3^+	$u_4 \to u_3$	$H_r = C_3^+ T G_4^+$	$N_-(H_r) \le 1$

For example, a left type-R vertex has side message V_2^- with q-weighted fourth moment at most one. The retained incoming edge contributes T, so (6.1) gives $N_+(H_\ell) \le 1$. A left type-S vertex has the fully reversed side message C_2^- with p^5-weighted fourth moment at most one; (6.4) gives the same N_+ output, using an outgoing outer path. The right type-R and type-S cases follow from (6.3) and (6.2), respectively.

The factor TG or AG integrates the final edge of the outer connector; its preceding $L-1$ edges are integrated in G. For an outer connector of length one, G is the zero-step terminal message.

The two side tables and the two terminal arm systems are disjoint, so their directions can be chosen independently. For each of the four role combinations RR, RS, SR and SS, the left half produces an N_+ bound and the right half an N_- bound.

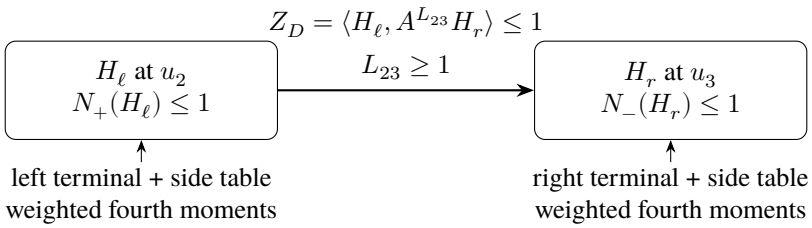


Figure 2: The chain merger. Terminal and side-table fourth moments give the two indicated half-tree bounds. The outer directions depend on the R/S roles in Table 6; the central connector always points from u_2 to u_3.

6.3 The complete central path

After integration over every arm and both outer connector interiors, the original density is exactly

$$Z_D(K,\mu) = \langle H_\ell, A^{L_{23}} H_r \rangle . \tag{6.5}$$

Lemma 2.1 gives

$$Z_D(K,\mu) \le \sqrt{N_+(H_\ell) N_-(H_r)} \le 1.$$

Together with the reducible-terminal case, this proves the theorem for the entire P_4 skeleton.

7 The star skeleton

Let o be the central branch vertex and u_1, u_2, u_3 the three terminal branch vertices, with connector lengths $L_1, L_2, L_3 \ge 1$. The center may have no side arms. Write E_c for its product of equal-length arm pairs, so $0 \le E_c \le 1$.

7.1 The center catalogue

The residual center tables fall into three classes. In the first, a fixed message C satisfies

$$\mathbb{E}[q^3C^4] \le 1. \tag{7.1}$$

The table gives the direction for each row. The factor E_c multiplies every row, and the products P and V^+ involve only the indicated residual arms.

Table 7: Center directions with a cubed-weight fourth moment.

Residual center table	Message C/E_c	Complete proof
Singleton $a \ge 1$	f_a	Section G.4
Odd, at least three, minimum one	pP, P canonical on the rest	Section K.1; (7.2)
Odd, at least three, minimum $a \ge 2$	f_aP, P canonical on the rest	Section K.1
Even, minimum two	$f_2h_{b_1}\prod_{i=2}^{t} h_{a_i}f_{b_i}$	Section K.3
Even, minimum three	$f_3h_{b_1}\prod_{i=2}^{t} h_{a_i}f_{b_i}$	Section K.2
Even, minimum at least four	$\prod_{i=1}^{t} h_{a_i}f_{b_i}$	Section K.2
Even, at least four, minimum one	pV^+	Section K.3; (7.2)

The unit-arm reductions in the second and last rows are direct. If $V \ge 0$ is the message of the row in question, with $\mathbb{E}[pV^4] \le 1$, then

$$\mathbb{E}[q^3(pV)^4] = \mathbb{E}[p(pq)^3V^4] \le \mathbb{E}[pV^4] \le 1. \tag{7.2}$$

For an odd center of minimum one, $V = P$ is the canonical even product. For an even center with at least four entries and minimum one, the remaining odd list has at least three entries, and V^+ is exactly its zero-step direction in (5.3). The leading p comes from the unit arm.

The singleton row directs its residual arm outward for every length and is proved in Section G.4, with a separate certificate for the two-edge arm.

The second center class consists of a residual list exactly $(1, b)$, $b \ge 2$. Direct both residual arms outward and take

$$C = E_cpf_b, \qquad \mathbb{E}[q^9C^4] \le \mathbb{E}[q^5f_b^4] \le 1. \tag{7.3}$$

Indeed $q^9C^4 = E_c^4(pq)^4q^5f_b^4$. The last inequality is the complete singleton estimate in Sections G.2, G.3. This class uses the ninth power rather than the cubed weight, including the case $b = 2$.

The third class has an empty residual list and message $C = E_c \le 1$; Section 7.4 handles it with a different merger.

The first class contains every odd nonempty list and every even list whose minimum is two, three, or at least four. An even list with minimum one has either two entries, which gives the second class, or at least four, which gives the first. The empty list is the third class.

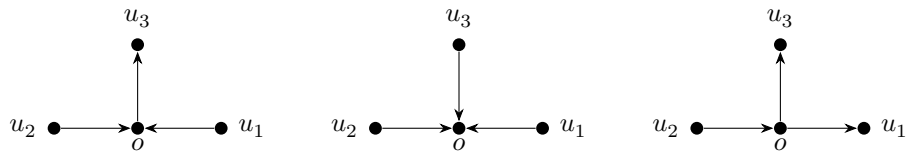
7.2 Cubed-weight centers: two incoming paths and one outgoing path

Assume (7.1). Reducible terminals have already been covered by Lemma 4.1. Fix the core directions

$$u_1 \longrightarrow o, \qquad u_2 \longrightarrow o, \qquad o \longrightarrow u_3,$$

with every original edge of each path pointing consistently. Put

$$G_i = T^{L_i-1}F^+_{i,L_i-1} \quad (i = 1, 2), \qquad H_3 = A^{L_3-1}F^-_{3,L_3-1}.$$



Cubed weight
$\mathbb{E}[q^3C^4] \le 1$

Ninth weight
$\mathbb{E}[q^9C^4] \le 1$

Empty residual center
$C = E_c \le 1$

Figure 3: Core directions for the three star-center constructions. The empty-center case uses a full L_1-step terminal on the right-hand path and a second-moment terminal on the upper path. All connector lengths remain positive and arbitrary.

The terminal theorem gives $\mathbb{E}[pG_i^4] \le 1$ and $\mathbb{E}[qH_3^4] \le 1$. Retaining only the final edge near o on each path, the density is

$$Z_D = \mathbb{E}[C(TG_1)(TG_2)(AH_3)]. \tag{7.4}$$

We control two groups at the same center label. First,

$$\begin{aligned} \|CTG_1\|_2^2 &\le \mathbb{E}[qC^2T(G_1^2)] \\ &\le \sqrt{\mathbb{E}[q^3C^4]\,\mathbb{E}[pG_1^4]} \le 1. \end{aligned} \tag{7.5}$$

The first step is row Cauchy–Schwarz; the second applies Cauchy–Schwarz to the T edge measure, whose root marginal is $q\mu$. For the other two paths, row Cauchy–Schwarz and $pq \le 1$ give

$$\begin{aligned} \|(TG_2)(AH_3)\|_2^2 &\le \mathbb{E}[pq\,T(G_2^2)A(H_3^2)] \\ &\le \langle G_2^2, A^2(H_3^2)\rangle \\ &\le \sqrt{\mathbb{E}[pG_2^4]\,\mathbb{E}[qH_3^4]} \le 1. \end{aligned} \tag{7.6}$$

The middle expression uses $T^* = A$, and the last step is Lemma 2.1 with length two. The joint estimate integrates the two adjacent edges at their common label. Applying ordinary Cauchy–Schwarz to the two groups in (7.4) proves $Z_D \le 1$.

7.3 Ninth-weight centers: all three paths incoming

Now use (7.3) and direct every connector from u_i toward o. Set $G_i = T^{L_i-1}F^+_{i,L_i-1}$, so $\mathbb{E}[pG_i^4] \le 1$. The exact density is

$$Z_D = \mathbb{E}\left[C\prod_{i=1}^{3} TG_i\right].$$

Hölder on a row of the nonnegative T kernel gives

$$(TG_i)^4 \le q^3T(G_i^4). \tag{7.7}$$

The factor q^3 reflects the mass q of the row measure. Apply (7.7), followed by four-factor Hölder on the original probability measure:

$$\begin{aligned} Z_D &\le \mathbb{E}\left[C q^{9/4} \prod_{i=1}^{3} T(G_i^4)^{1/4} \right] \\ &\le \left(\mathbb{E}[q^9 C^4] \prod_{i=1}^{3} \mathbb{E}[T(G_i^4)] \right)^{1/4} \\ &= \left(\mathbb{E}[q^9 C^4] \prod_{i=1}^{3} \mathbb{E}[pG_i^4] \right)^{1/4} \le 1. \end{aligned}$$

The equality uses $\mathbb{E}[Tv] = \mathbb{E}[pv]$.

7.4 Empty residual center

Here $C = E_c \le 1$. Choose the core directions $o \to u_1$, $u_2 \to o$, and $o \to u_3$. At the first terminal use the *positive-step* supply through the entire first path:

$$X = A^{L_1} F^{-}_{1,L_1}, \qquad \mathbb{E}[qX^4] \le 1.$$

At the second terminal retain its last edge and put

$$G = T^{L_2-1} F^{+}_{2,L_2-1}, \qquad \mathbb{E}[pG^4] \le 1.$$

At the third terminal choose the actual spider direction of Lemma 2.2, with message F_3 and $N_-(F_3) \le 1$; a second-moment bound suffices here.

Let $H = X(TG)$. Row Cauchy–Schwarz, followed by edge Cauchy–Schwarz with the correct marginals, yields

$$\begin{aligned} N_+(H) &\le \mathbb{E}[pqX^2 T(G^2)] \\ &\le \sqrt{\mathbb{E}[p^2 q^3 X^4]\, \mathbb{E}[pG^4]} \\ &\le \sqrt{\mathbb{E}[qX^4]\, \mathbb{E}[pG^4]} \le 1. \end{aligned}$$

In the last comparison $p^2 q^2 \le 1$ is used. Multiplying by $C \le 1$ gives $N_+(CH) \le 1$. The exact density is

$$Z_D = \langle CH, A^{L_3} F_3 \rangle \le \sqrt{N_+(CH) N_-(F_3)} \le 1.$$

The three center constructions, the terminal catalogue, and Lemma 4.1 prove the assertion for the $K_{1,3}$ skeleton.

8 Five branch vertices: reduction and geometry

Let B be the five original branch vertices and let S be their minimal connecting subtree. Its leaves belong to B, and every vertex of degree at least three in S belongs to B. Suppressing the degree-two vertices outside B reveals the skeleton. The resulting tree on five vertices is P_5, $K_{1,4}$ or T_5: maximum degree four gives the star, maximum degree three gives degree sequence $(3, 2, 1, 1, 1)$, and maximum

degree two gives the path. Each skeleton edge retains its original positive path length, and every off-core component is a side arm.

At each skeleton vertex v, pair equal-length side arms maximally. Let c_v be its skeleton degree and t_v the number of remaining distinct arm lengths.

Lemma 8.1. *If $c_v + t_v \le 2$ at any skeleton vertex, then U has an orientation satisfying (1.3) for every finite model.*

Proof. Orient each equal pair as in (2.5), then bound their product at v by one in the integrand. The comparison tree retains v, all core paths and the unpaired arms, and has at most four branch vertices; Theorem 3.2 gives it a fixed orientation. Restoring the pairs at their original attachment multiplies the integrand at v by their actual product, which is at most one, so the original density is also at most one. □

Apply this test at every skeleton vertex. In P_5 it covers an empty internal residual and a terminal with at most one residual arm; an empty residual at the star or fork center needs the separate argument of Section 10.

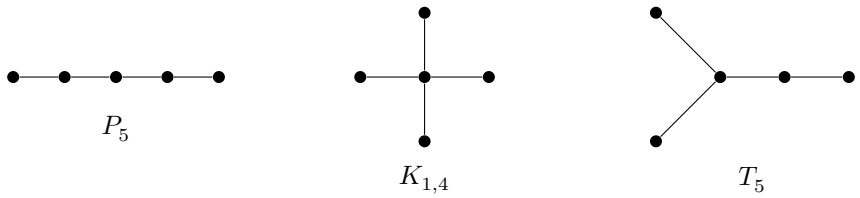


Figure 4: The three skeletons on five branch vertices. Every drawn edge denotes an independently chosen positive-length path in the original tree.

9 The five-branch boundary estimates

Besides $f_n, \bar{f}_n, h_n$, write

$$W_n = A\bar{f}_{n-1} \quad (n \ge 1), \qquad I_q(V) = \mathbb{E}[q^{-1/3}|V|^{8/3}].$$

The arm message W_n directs its first edge away from the root and its remaining edges toward the first neighbor. Reversal exchanges f_n with $\bar{f}_n$ and h_n with W_n.

9.1 Simple-reference and ordinary moments

The terminal residuals below contain at least two distinct lengths. All messages include their actual equal-pair factors.

Proposition 9.1 (Simple-reference fourth moments)**.** *For every existing outward arm f_s, $s \ge 1$, every terminal arm multiset as above, and every connector length $L \ge 1$, there is a structurally chosen terminal direction with actual received message $X = T^L F$ such that*

$$\mathbb{E}[f_s X^4] \le 1.$$

For $s \ge 3$, the direction in the genuine-reference table satisfies the stronger bound $1 - d/3000$; for $s = 2$ only the bound 1 is used.

For $s = 1$ the assertion is Theorem 5.1 with step count L. For $s \geq 2$ the direction table and its proof are in Sections N.41 and N.42, indexed in Appendix L; the short-arm directions there are specific to this reference catalogue.

For ordinary moments, a *post block* means actual unequal central pairs with negative canonical message

$$P = \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}, \qquad 4 \leq u_1 < v_1 < \cdots < u_k < v_k.$$

When nonempty, it is required that the designated anchor length satisfy $m < u_1$. This order is automatic in the central allocations below.

Proposition 9.2 (Ordinary eighth moments)**.** *Given an existing arm of length $m \geq 3$, a terminal residual of size at least two, a connector of length $L \geq 1$ and a post block in the stated order, there is a direction of exactly these resources with actual message H satisfying*

$$\mathbb{E}H^8 \leq e^{-d/1024}.$$

Given instead one existing unit arm and no unequal post block, there is an actual message H_1 with $\mathbb{E}H_1^8 \leq 1 - d/20000$.

The complete ordinary direction table, including the first-edge choices, is in Section O.39, and Appendix L indexes its cases. The nonunit proof draws on Sections O.32, O.34, O.35, O.36 and O.38; the unit input is the ordinary catalogue of Section O.5. For the terminal $(1, 2, 3)$, the special direction occurs at $L = 2$ in this ordinary catalogue and at $L = 1$ in the genuine-reference catalogue.

We also use the following path and mixed central bounds:

$$\mathbb{E}f_s^2 \leq 1 \quad (s \geq 2), \qquad \mathbb{E}[f_s^2 Q^8] \leq e^{-17d/64}, \tag{9.1}$$

where the second bound has $Q = W_a \bar{f}_b \prod_i W_{u_i} \bar{f}_{v_i}$ and $2 \leq s < a < b < u_1 < v_1 < \cdots$. The mixed proof, including $s = 2$ and $a = 3$, is in Section R.4. When $Q = 1$ only the first bound is needed. For $s = 2$ its exact value is $1 - d + \delta_2$; for $s \geq 3$ the finite path estimate gives $\mathbb{E}f_s^2 \leq 1 - d + 2\delta_2 \leq 1 - 3d/16$.

9.2 A reference second moment for a compound branch

A fork has a two-segment branch $c - r - t$ with independent lengths $M, N \geq 1$ and arbitrary side-arm tables at r, t.

Proposition 9.3. *For each $s \geq 1$, this actual compound branch admits a direction with received message Y satisfying $\mathbb{E}[f_s Y^2] \leq 1$.*

Proof. If t is reducible, its retained connector and possible residual arm form one comparison arm at r. Choose the canonical endpoint direction used in (C.9), apply that inequality, and restore the equal pairs. Otherwise put $G = T^{N-1} F^{+}_{t,N-1}$, so $\mathbb{E}[pG^4] \leq 1$.

If the residual at r is empty or has at least two entries, select its negative zero-step factor C_r^- with $\mathbb{E}[q(C_r^-)^4] \leq 1$ and set $H = C_r^- TG$, $Y = T^M H$. Row Hölder and Cauchy–Schwarz give

$$(TG)^4 \leq q^3 T(G^4), \qquad \mathbb{E}[H^2/q] \leq \sqrt{\mathbb{E}[q(C_r^-)^4]\,\mathbb{E}[(TG)^4/q^3]} \leq 1. \tag{9.2}$$

Thus $H = \sqrt{q}\, h$ with $\|h\|_2 \leq 1$. The operator bounds in Proposition C.4, together with normalized propagation, give

$$\sqrt{f_s} T^M \sqrt{q} = (\sqrt{f_s} T \sqrt{q})(q^{-1/2} T \sqrt{q})^{M-1}, \qquad \|\sqrt{f_s} T^M \sqrt{q}\| \leq 1. \tag{9.3}$$

This proves the claim in the R case, including $M = N = 1$.

If r has one residual arm of length a, orient it inward as $\bar{f}_a$. At t, choose the positive canonical direction for an even residual list and the negative canonical direction for an odd residual list, and denote the resulting message by S. Let S^- be its full reversal and put $V = A^N S^-$, $H = \bar{f}_a V$ and $Y = T^M H$. The full reversal of (C.9) gives $\mathbb{E}[\bar{f}_a V^2] \le 1$ for every $a, N \ge 1$. The cross-path contraction $\|\sqrt{f_s} T^M \sqrt{\bar{f}_a}\| \le 1$ proves the assertion. Equal pairs at r are restored by positivity. □

9.3 The common-reference outlets for the path skeleton

Consider one side $c - r - t$ of P_5. Its terminal connector $r - t$ has length $L \ge 1$, and its receiving connector $r - c$ has length $M \ge 1$. The terminal residual has size at least two, and the residual at r is nonempty. The following catalogue is used after the two whole-tree alternatives of Section 11.

Proposition 9.4 (Common-reference outlet)**.** *Except for a unit singleton at r with $M = 1$, every such side has an actual message U at c, determined by the structure, satisfying*

$$I_q(U)^3 \le e^{-\eta d}, \qquad \eta \ge 1/500. \tag{9.4}$$

Both sides of the path use the same q.

Table 8: The path outlet catalogue. The entries locate the proofs in the technical appendix, and Appendix L indexes the direction tables.

Root residual	Structural subdivision	Where proved
Unit singleton	$M = 2$; $M \ge 3$	P.4, P.5, P.6, P.7, P.8, P.9; M.19, O.5
Singleton $a \ge 2$	$M = 1, 3, \ge 4$; $a = 2, M = 2$; $a \ge 3, M = 2$	N.41, N.42, R.5, R.1; R.6; R.7
Exactly two arms	Minimum one; minimum at least two	M.16, O.5; O.39
Odd, at least three	Minimum one, two, or at least three	Q.13, Q.21, Q.22, Q.23, Q.24; Q.25–Q.28
Even, at least four	Minimum one or at least two, with one actual inward multiplier retained	R.3

The unit-singleton row $M = 1$ is handled by the whole-tree construction before this catalogue is consulted.

10 The five-vertex star and fork

Let c be the degree-four star center or the degree-three fork center. There are four simple terminal branches in the star, and two simple terminal branches plus one compound branch in the fork. A reducible simple terminal is covered by Lemma 8.1. Write $E_c \le 1$ for the product of central equal pairs.

10.1 Nonempty central residuals

Table 9 allocates the central arms along the increasing residual list. One actual outward arm f_s serves as the common reference; an ordinary message consumes its anchor, one simple terminal and the connecting path; the central equal pairs belong to one indicated factor. In every nonsingleton row $s \ge 2$,

Table 9: Central arm allocation for $K_{1,4}$ and T_5. Later pairs always retain their displayed negative canonical directions.

Residual list	Reference	Ordinary factors	Other central factors
Singleton s	f_s	None	E_c in one reference message
$(s, m, \text{later pairs}), 2 \le s < m$	f_s	H: anchor m, later pairs, E_c	None
$(s, a, b, \text{later pairs}), 2 \le s < a < b$	f_s	H_a and H_b; latter holds later pairs and E_c	None
$(1, s, m, \text{later pairs})$	f_s	Unit H_1; H_m holds later pairs and E_c	None
$(1, s)$ or $(1, s, a, b, \text{later pairs})$	f_s	Unit H_1 holds E_c	$Q = 1$ or $W_a \bar{f}_b$ times later pairs

so (9.1) applies, and the anchor preceding a nonempty post block is shorter than all its post lengths, as Proposition 9.2 requires.

For a singleton center the densities are

$$Z_\star = \mathbb{E}[f_s X_1 X_2 X_3 X_4], \qquad Z_{T_5} = \mathbb{E}[f_s X_1 X_2 Y].$$

Use Hölder exponents $(4, 4, 4, 4)$ or $(4, 4, 2)$ and the reference estimates. Both densities are at most one, including $s = 1, 2$.

For a nonunit even center the products are $f_s H X_2 X_3 X_4$ and $f_s H X Y$. Factor the first into $H, f_s^{1/4}, f_s^{1/4} X_2, f_s^{1/4} X_3, f_s^{1/4} X_4$, and the second into $H, f_s^{1/4}, f_s^{1/4} X, f_s^{1/2} Y$. Hölder with exponents $(8, 8, 4, 4, 4)$ or $(8, 8, 4, 4, 2)$ gives, in both cases, $Z_D \le e^{-d/8192}$.

For a nonunit odd center there are two distinct ordinary factors. The star and fork factors are, respectively,

$$(H_a, H_b, f_s^{1/2}, f_s^{1/4} X_3, f_s^{1/4} X_4), \qquad (H_a, H_b, f_s^{1/2}, f_s^{1/2} Y).$$

The same exponent lists give $Z_D \le e^{-d/4096}$. For a unit-minimum odd center replace the first ordinary input by $\mathbb{E} H_1^8 \le 1 - d/20000$. This gives

$$Z_D \le (1 - d/20000)^{1/8} e^{-d/8192} \le e^{-(1/160000 + 1/8192) d}.$$

For a unit-minimum even center use the factors $H_1, f_s^{1/4} Q$ and the remaining reference factors. Equations (9.1) and the same Hölder exponents give

$$Z_D \le \begin{cases} e^{-d/160000}, & Q = 1, \\ e^{-(1/160000 + 17/512) d}, & Q \ne 1. \end{cases} \tag{10.1}$$

The reference arm carries total analytic power one, and the ordinary factors use distinct terminal branches. Hölder applies to the resulting functions of the central label.

10.2 Empty central residual

For the star, choose two paths toward c and two away from it. Put $G_i = T^{L_i - 1} F^+_{i, L_i - 1}$ for $i = 1, 2$ and $H_i = A^{L_i - 1} F^-_{i, L_i - 1}$ for $i = 3, 4$. Then

$$Z_D = \mathbb{E}[E_c (T G_1)(T G_2)(A H_3)(A H_4)].$$

For each pair, row Cauchy–Schwarz, $pq \le 1$, and the path interface give

$$\mathbb{E}[(TG)^2(AH)^2] \le \langle G^2, A^2(H^2)\rangle \le \sqrt{\mathbb{E}[pG^4]\,\mathbb{E}[qH^4]} \le 1.$$

Cauchy–Schwarz on $(TG_1)(AH_3)$ and $(TG_2)(AH_4)$ proves $Z_D \le 1$.

For the fork label its simple ends a, b and its compound $c - r - t$, with lengths L_a, L_b, M, N. Set

$$X = A^{L_a} F^-_{a,L_a}, \quad G = T^{L_b-1} F^+_{b,L_b-1}, \quad H_c = E_c X(TG).$$

The first terminal uses the L_a-step supply. The edge measure for T has marginals $q\mu, p\mu$, so

$$N_+(H_c) \le \sqrt{\mathbb{E}[p^2q^3X^4]\,\mathbb{E}[pG^4]} \le \sqrt{\mathbb{E}[qX^4]\,\mathbb{E}[pG^4]} \le 1.$$

If r is type R, choose its positive zero-step factor V_r^+ and put $V = A^{N-1}F^-_{t,N-1}$, $H_r = V_r^+ AV$. If r is a singleton, put $C_r^+ = E_r f_a$, $V = T^{N-1}F^+_{t,N-1}$ and $H_r = C_r^+ TV$. Inequality (6.3) gives $N_-(H_r) \le 1$ in the R case, and (6.2) in the singleton case, where the q^5 weight enters. Direct the whole M-path from c to r. Then

$$Z_D = \langle H_c, A^M H_r\rangle \le \sqrt{N_+(H_c)N_-(H_r)} \le 1.$$

This proves both skeletons. Only the nonsingleton central allocations give the strict loss $e^{-d/160000}$; the comparison, empty-center and singleton-center directions give the bound 1.

11 The five-vertex path

Write the core in path order as

$$t \;{-}{-}_{L_-}\; r \;{-}{-}_{M_-}\; c \;{-}{-}_{M_+}\; w \;{-}{-}_{L_+}\; z, \qquad L_-, M_-, M_+, L_+ \ge 1.$$

Here $L_\pm$ counts a terminal connector and $M_\pm$ the connector from an off-center root to c. In a local outlet formula, L denotes the terminal-connector length and M the receiving-connector length. After Lemma 8.1, the two terminal residuals have at least two entries and the three internal residuals are nonempty. Call an internal residual S if it is a singleton and R otherwise.

11.1 A central singleton: the maximal S spine

Suppose c is S. Take the maximal consecutive S sequence among r, c, w that contains c. Orient its singleton arms alternately as f_a and $\bar{f}_b$, starting with an outward arm at its left end. Every connector between consecutive S vertices points from the negative end toward the positive end. The corresponding operators between square roots of the actual arm messages are

$$B_i = \sqrt{f_a}\,T^{M_i}\sqrt{\bar{f}_b} \quad \text{or their adjoints}, \qquad \|B_i\| \le 1$$

by Proposition C.4.

At a positive S end whose arm has message f_s, an ordinary terminal supplies

$$y = \sqrt{f_s}\,T^L F_t, \qquad \|y\|_2 \le 1,$$

where F_t is positive canonical for an even residual list and negative canonical for an odd residual list; the norm bound is (C.9). An R half-tree supplies instead

$$y = \sqrt{f_s}\,T^M H, \qquad H = C^- TG = \sqrt{q}\,h, \qquad \|h\|_2 \le 1.$$

Equations (9.2) and (9.3) give $\|y\|_2 \le 1$ for every $M \ge 1$. At a negative S end reverse the entire outside configuration: the two endpoint functions become $\sqrt{\bar{f}_s}A^L F_t^-$ and $\sqrt{\bar{f}_s}A^M H^-$, respectively, with the same norm bound.

Bounding the genuine equal-pair factors on the spine by one gives

$$\langle y_L, B_1 \cdots B_{j-1} y_R \rangle \le 1, \qquad \|y_L\|_2, \|y_R\|_2 \le 1.$$

For a one-vertex spine this is $\langle y_L, y_R \rangle$. This covers the patterns SSS, RSS, SSR and RSR. The two square roots of each arm contribute one arm factor, and each original connector appears once.

11.2 A unit off-center arm with a unit receiving connector

Now suppose the center is R and, say, r has one unit residual arm and its receiving connector $r-c$ has $M_- = 1$. Put $L = L_-$ and choose the terminal direction with step count $L-1$. Its composite message at r is

$$F_*^+ = E_r^- q A^L F_{t,L-1}^-.$$

The actual unit arm is the factor q. Row Hölder and $pq \le 1$ give, for every real G,

$$\mathbb{E}[q|pTG|^4] \le \mathbb{E}[(pq)^4 T(|G|^4)] \le \mathbb{E}[p|G|^4].$$

Full reversal proves $\mathbb{E}[p(F_*^+)^4] \le 1$. The one edge from r to c has not yet been integrated. Choose the central negative R factor C^- with $\mathbb{E}[q(C^-)^4] \le 1$ and put $H_c = C^- T F_*^+$. Lemma 6.1 gives $N_+(H_c) \le 1$.

At w, use the right R or S construction in Table 6, with its actual terminal and terminal-connector length L_+. It gives a message H_w with $N_-(H_w) \le 1$. Direct the entire M_+-path from c to w. The exact density is

$$Z_D = \langle H_c, A^{M_+} H_w \rangle \le 1.$$

This whole-tree construction replaces the I_q outlet at r.

11.3 Two compatible outlets

In the remaining structural branch c is R, and any unit singleton off-center has receiving-connector length at least two. Choose the central negative zero-step factor C_c as follows. An even residual uses negative canonical pairs. An odd residual beginning with one uses $E_c^- p \bar{f}_a W_b$ followed by negative canonical pairs. A nonunit odd residual uses $E_c^+ f_a$ followed by positive canonical pairs. These are the full reversals of the corresponding zero-step directions, so

$$\mathbb{E}[qC_c^4] \le 1.$$

Proposition 9.4 supplies actual outputs U_-, U_+ for the two sides, both using this same q. Integration over their disjoint interiors gives $Z_D = \mathbb{E}[C_c U_- U_+]$. Hölder at exponents $4, 8/3, 8/3$ yields

$$\begin{aligned} Z_D &\le (\mathbb{E}[qC_c^4])^{1/4} I_q(U_-)^{3/8} I_q(U_+)^{3/8} \\ &\le e^{-(\eta_- + \eta_+)d/8} \le e^{-d/2000} \le 1. \end{aligned} \tag{11.1}$$

The weight powers $1/4, -1/8, -1/8$ cancel, and both side functions are evaluated at the same central label.

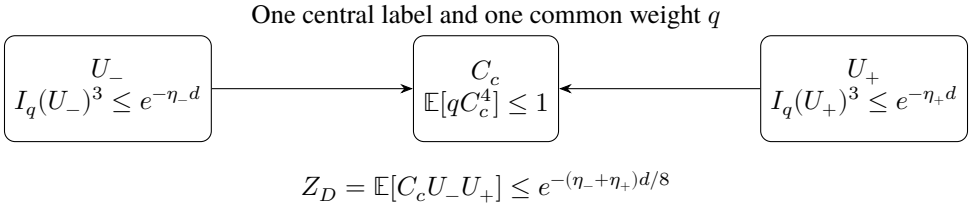


Figure 5: The remaining path branch. The two structural alternatives are applied first; only then are the two local outlets joined with the same central reference weight.

The root catalogue divides nonempty residual lists into singletons, pairs, odd lists of at least three entries, and even lists of at least four. Singletons split according to their arm and receiving-connector lengths: the two-step construction treats an arm of length two differently from one of length at least three, and the choice depends on the length; d enters in the estimates. The minimum-two odd row with a terminal $(2, t)$ has its own first-edge choices, notably $W_2\bar{f}_t$ at $L = 1$ and $\bar{f}_2 W_t$ at $L \ge 2$; the table in the technical appendix lists them all. Together with the two whole-tree alternatives above, (11.1) proves every P_5.

11.4 A worked five-branch example

Consider the path core in Figure 6, with connector lengths $(L_-, M_-, M_+, L_+) = (2, 1, 2, 1)$ and pendant-arm lists

vertex	t	r	c	w	z
arm lengths	$(2, 4)$	$(1, 3, 4)$	$(4, 5)$	$(2, 4, 5)$	$(2, 3)$.

(11.2)

This tree has 45 edges and five branch vertices, of degrees $3, 5, 4, 5, 3$. It falls in the two-outlet case above. Direct every core edge toward c, and choose the pendant directions specified by the following messages:

$$\begin{aligned} F_- &= \bar{f}_2 W_4, \quad H_- = pW_3\bar{f}_4\, T^2 F_-, \quad U_- = TH_-, \\ F_+ &= W_2\bar{f}_3, \quad H_+ = f_2 h_4 f_5\, TF_+, \quad U_+ = T^2 H_+, \\ &\qquad C = W_4\bar{f}_5. \end{aligned} \tag{11.3}$$

Integration over the two sides gives $Z_D = \mathbb{E}[CU_-U_+]$. We verify the three moment inputs to this identity directly. When $d = 0$, leaf integration gives $Z_D = 1$; henceforth assume $d > 0$.

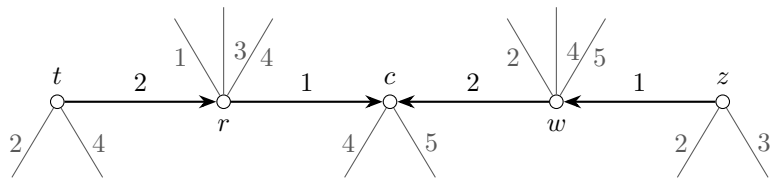


Figure 6: The five-branch example. Each segment represents a path of the indicated length; pendant directions are specified in (11.3).

Put $B = -K$ and $u = -g = B\mathbf{1}$, so that $T = J + B$, $A = J - B$, $p = 1 - u$, and $q = 1 + u$. We use $d \le 1/3$, $\delta_{j+1} \le \theta\delta_j$, $\theta = 13/32$, and the spectral measure ν of $-B^2$ at u, for which $\int x^j\, d\nu = \delta_{j+1}$. The row bound $|BV| \le \|V\|_2$ and (M.43) give the common fourth-moment rule

$$0 \le H \le e^{a+BV} \quad \Longrightarrow \quad \log \mathbb{E}H^4 \le 4(a + \mathbb{E}BV) + \tfrac{17}{4}\|V\|_2^2. \tag{11.4}$$

The arm recurrences $f_{n+1} = Af_n$, $\bar{f}_{n+1} = T\bar{f}_n$, $h_n = Tf_{n-1}$, and $W_n = A\bar{f}_{n-1}$ give

$$\begin{aligned} \bar{f}_2 + W_4 &= 2 - 2d + du - B^2u - B^3u, \\ W_3 + \bar{f}_4 &= 2 - 3d - du + B^3u, \\ W_2 + \bar{f}_3 &= 2 - d + B^2u, \\ h_4 + f_5 &= 2 - 4d + \delta_2 - df_2 - B^4u, \\ W_4 + \bar{f}_5 &= 2 - 4d + \delta_2 - d\bar{f}_2 + B^4u. \end{aligned} \tag{11.5}$$

These identities and $ab \le e^{a+b-2}$ for nonnegative a, b provide all the envelopes below.

The two-step side.

Set

$$R = du - B^2u - B^3u, \quad s = \mathbb{E}R^2 = d^3 + 2d\delta_2 + \delta_3 + \delta_4, \quad \Delta = -\delta_2 + \tfrac{3}{2}s, \quad \epsilon = \Delta - d^2 - \delta_2.$$

The first identity gives $F_- \le e^{-2d+R}$. Since $|R| \le d + \sqrt{\delta_2 + \delta_3} < 1$, the elementary bound $e^y \le 1 + y + 3y^2/4$, $|y| \le 1$, and positivity give

$$TF_- \le e^{-2d}(q + BR + \Delta), \qquad T^2F_- \le e^{-2d}(\bar{f}_2 + B^2R + \Delta q - d^2 - \delta_2).$$

Adding the factors in the last upper bound to p, and then using the second identity in (11.5), yields

$$H_- \le e^{-5d+\epsilon+BV_-}, \qquad V_- = u + BR + B^2u + (\Delta - d)\mathbf{1}.$$

Here

$$-5d + \epsilon + \mathbb{E}BV_- = -6d + \epsilon + \delta_2 + \delta_3 < -\tfrac{27}{5}d.$$

Indeed

$$\epsilon = d^2(-1 + 3d/2) + (-2 + 3d)\delta_2 + \tfrac{3}{2}(\delta_3 + \delta_4) \le 0,$$

because $3\theta(1+\theta)/2 < 1$.

To bound the complete vector, write $c_0 = \Delta - d$. Its nonconstant part has spectral coordinates $a(x) = 1 - x - x^2$, $b(x) = d + x$. On $[0, \theta]$, $a^2 \le 1 - x$ and $0 \le b \le 1$, so that part has squared norm at most d and mean $-d^2 - \delta_2$. Moreover

$$s/d \le \tfrac{1}{9} + \tfrac{2\theta}{3} + \theta^2 + \theta^3 < \tfrac{5}{8}, \qquad -(d + \delta_2) \le c_0 < 0.$$

Keeping its constant cross term therefore gives

$$\|V_-\|_2^2 \le d + (d + \delta_2)^2 + 2(d + \delta_2)(d^2 + \delta_2) < \tfrac{12}{5}d.$$

The rule (11.4) now proves $\log \mathbb{E}H_-^4 < -57d/5 \le -6d$.

The one-step side.

The third identity gives $F_+ \le e^{-d+B^2u}$. The same Taylor bound gives

$$TF_+ \le e^{-d}(q + B^3u + \tfrac{3}{2}\delta_3).$$

Adding its upper factor to $f_2 = 1 - u + Bu$ leaves $2 + B(u + B^2u) + 3\delta_3/2$. The fourth identity, together with $f_2 \geq 0$ and $4 - \theta > 3$, gives $h_4 f_5 \leq e^{-3d - B^4 u}$. Consequently

$$H_+ \leq e^{-4d + (3/2)\delta_3 + BV_+}, \qquad V_+ = u + B^2u - B^3u.$$

Its spectral square is $(1-x)^2 + x^3 \leq 1$. Thus

$$\|V_+\|_2^2 \leq d, \qquad -4d + \tfrac{3}{2}\delta_3 + \mathbb{E}BV_+ = -5d + \delta_2 + \tfrac{3}{2}\delta_3 < -4d.$$

Equation (11.4) proves $\log \mathbb{E}H_+^4 < -47d/4 \leq -6d$.

The center and the receiving connectors.

The last identity gives $C \leq e^{-3d + B^4 u}$. Put $X = u + 4B^4 u$. Skew-adjointness gives $\mathbb{E}X = 0$, and

$$|X| \leq 1 + 4\theta^{3/2}\sqrt{d} < 2, \qquad \mathbb{E}X^2 \leq (1 + 4\theta^2)^2 d < 3d.$$

Since $q \leq e^u$ and $e^y \leq 1 + y + 3y^2/2$ for $|y| \leq 2$,

$$\mathbb{E}[qC^4] \leq e^{-12d}\mathbb{E}e^X \leq e^{-12d}(1 + 9d/2) \leq e^{-15d/2} \leq 1.$$

For a positive operator P, row Cauchy–Schwarz followed by Hölder gives

$$I_q(PH)^3 \leq \mathbb{E}[(P\mathbf{1})^4/q]\, \|P\|_{2\to 2}^4 (\mathbb{E}H^4)^2.$$

The short-path row and operator bounds proved in Section Q.1 are

$$\mathbb{E}[q^3] \leq e^{4d}, \quad \|T\|^4 \leq e^{128d/19}, \qquad \mathbb{E}[\bar{f}_2^4/q] \leq e^{6d}, \quad \|T^2\|^4 \leq e^{4d}.$$

Using $\mathbb{E}H_-^4, \mathbb{E}H_+^4 \leq e^{-6d}$ gives

$$I_q(U_-)^3 \leq e^{-24d/19}, \qquad I_q(U_+)^3 \leq e^{-2d}.$$

Finally, the three-factor Hölder inequality at c yields

$$Z_D \leq (\mathbb{E}[qC^4])^{1/4} I_q(U_-)^{3/8} I_q(U_+)^{3/8} \leq e^{-(24/19+2)d/8} \leq 1.$$

The powers $q^{1/4}, q^{-1/8}, q^{-1/8}$ cancel at the central label.

12 The fixed orientation and its consequences

Fix an ordering of the original vertices and arms to resolve ties. For $b(U) \leq 4$ use Theorem 3.2. Otherwise recognize the five-vertex skeleton and apply Lemma 8.1 at the first qualifying vertex, if one exists. For $K_{1,4}$ or T_5, use the corresponding central allocation or empty-center construction of Section 10. For P_5, first use the maximal S spine when the center is S, then the construction for a unit arm with a receiving connector of length one when applicable, and otherwise select the two outlet rows from the original integer data. Restore every previously oriented equal pair at its original attachment.

This rule fixes one orientation of U before the host is chosen and proves Theorem 1.2.

12.1 All maps into a tournament

For a loopless tournament $\mathcal{T}$ on $n_T \geq 1$ vertices, take the uniform measure and its skew sign kernel, with $K(x,x) = 0$. For every original edge and every vertex map ϕ,

$$\mathbf{1}_{\{\phi(u)\to\phi(v) \text{ in } \mathcal{T}\}} \leq \frac{1 + K(\phi(u),\phi(v))}{2}.$$

For distinct images this is equality; for equal images it is the inequality $0 \leq 1/2$. Multiplying over the original edges and summing over all $n_T^{|V(U)|}$ maps gives

$$\hom(D,\mathcal{T}) \leq 2^{-e(U)} n_T^{|V(U)|} Z_D(K,\mu) \leq 2^{-e(U)} n_T^{e(U)+1},$$

which proves Theorem 1.1.

12.2 Correlated vertex labels

Let W_D be the product in (1.2) and let $P = \mu^{V(U)}$. A constant-label assignment has positive mass and weight one, so $Z_D > 0$.

Corollary 12.1. *For the orientation in Theorem 1.2 and every joint law ρ on the original vertex labels,*

$$\mathbb{E}_\rho \log W_D \leq D_{\mathrm{KL}}(\rho\|P) + \log Z_D \leq D_{\mathrm{KL}}(\rho\|P).$$

Proof. When the reference entropy is finite and ρ gives no mass to $W_D = 0$, use the probability law $dP_D = (W_D/Z_D)dP$. Nonnegativity of $D_{\mathrm{KL}}(\rho\|P_D)$ gives the first inequality, and the kernel theorem gives the second. If ρ gives positive mass to $W_D = 0$, the left side is $-\infty$. A law outside the reference support has infinite reference entropy and is covered by the extended-value form of the inequality. □

12.3 Boundary cases

Zero-mass atoms can be omitted without changing the integral. At $d = 0$ we have $K\mathbf{1} = 0$, hence $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$, and successive leaf integration gives $Z_D = 1$ for every oriented tree. Reversing every original edge amounts to replacing K by $-K$, which preserves the universal bound.

13 Further directions

The four-branch argument converts terminal fourth moments into weighted second moments. For five branch vertices, the proof allocates central arms to different moment estimates and introduces additional boundary quantities, among them I_q. Each skeleton has a separate assembly. An extension to more branch vertices requires estimates that remain valid for the composite messages produced by further mergers.

Known attachment constructions already produce families with unbounded branch count [6, Theorem 1.3 and Proposition 5.2]. Attaching two copies with the same internal orientation and rooted message $m \geq 0$, where $\mathbb{E}m \leq 1$, through oppositely directed attachment edges contributes $(Am)(Tm) = (\mathbb{E}m)^2 - (Km)^2 \leq 1$, the kernel form of those constructions. Asymmetric composite branches require further boundary estimates.

Two branches give explicit losses, $e^{-d/2000}$ for the two-outlet path construction and $e^{-d/160000}$ for the nonsingleton central allocations. The other constructions yield $Z_D \leq 1$.

Author contributions

Both authors contributed equally, share first authorship, and are corresponding authors.

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Conflicts of interest

The authors declare no conflicts of interest.

Data and code availability

No experimental data were collected. Proof sources, exact-arithmetic verification scripts and rational certificate data are publicly available in the research compendium [7], GitHub release v2026.10.30-r3.

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AI assistance was used during symbolic exploration and manuscript preparation: OpenAI Codex assisted with merging, translation and typesetting; DeepSeek assisted with language editing and with language polishing and typesetting of the final manuscript. The authors are responsible for the content of this manuscript.

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